cts.tcl

```
# Description  : Clock Tree Synthesis (CTS) script.
# Tool         : Synopsys ICC2 (or similar implementation tool)
#
# Requirements :
#   - tech_setup.tcl, user_setup.tcl
#   - ${design}_place_done block already created
#
# Key Variables (from user_setup.tcl):
#   max_core                 - Max CPU cores
#   ndm_lib_name             - Library name under db/
#   design                   - Top-level design name
#   min_route_layer          - Min routing layer for CTS
#   max_route_layer          - Max routing layer for CTS
#   root_level_clk_route_min_layer / max_layer
#   leaf_level_clk_route_min_layer / max_layer
#   clk_tran_target          - Clock transition target
#   insertion_delay_target   - Clock insertion delay target
#   skew_target              - Clock skew target
#   flipsinv40c              - Corner set for CTS
#   clock_cells              - List of allowed CTS cells (BUF/INV etc.)
#   enable_ccd_cts           - Enable CCD during CTS (0/1)
#   cts_cong_effort          - (optional) CTS congestion effort
#   scts_hold_fix_effort     - (optional) CTS hold fix effort
#   setup_uncertainty_cts    - % of clock period used as setup uncertainty
#   hold_uncertainty_cts     - Absolute hold uncertainty
```

```
#   stop_after_cts            - If 1, stop after CTS

# Read Technology & User Settings

source -e -v ./scripts/tech_setup.tcl
source -e -v ./scripts/user_setup.tcl

# Multi-CPU Setting
set_host_options -max_cores $max_core

# Open NDM Library and Create CTS Block
if { [file exists "db/${ndm_lib_name}"] } {
   puts "\n[INFO] Opening NDM library 'db/${ndm_lib_name}'...\n"
   open_lib db/${ndm_lib_name}
   copy_block -from ${design}_place_done -to ${design}_cts_done
   current_block ${design}_cts_done
   link
} else {
   puts "\n[ERROR] NDM library 'db/${ndm_lib_name}' does not exist.
Please check.\n"
   return
}

# Define VT Types
puts "\n[INFO] Setting VT groups for LVT/RVT/MVT.\n"
```

```
set_attribute -quiet [get_lib_cells -quiet */*MVT*] threshold_voltage_group
MVT

set_attribute -quiet [get_lib_cells -quiet */*LVT*] threshold_voltage_group
LVT

set_attribute -quiet [get_lib_cells -quiet */*RVT*] threshold_voltage_group
RVT


set_threshold_voltage_group_type -type low_vt    LVT

set_threshold_voltage_group_type -type normal_vt RVT

set_threshold_voltage_group_type -type high_vt   MVT


# Basic Clock-Tree Check (pre-CTS)

set rpt_dir ./rpts/cts

if { ![file exists $rpt_dir] } {

   file mkdir $rpt_dir

}


puts "\n[INFO] Running pre-CTS clock-tree checks.\n"

check_clock_tree > $rpt_dir/check_clock_tree.rpt

# Routing Layer Settings for CTS

puts "\n[INFO] Setting CTS routing layer range: $min_route_
```